Agentic DataViz

Build Systems, Not Charts

A chart is one output.

The useful thing is the process that can make a better chart next time.

VizChitra 2026 · 3-hour workshop

Today

1. What makes a visualisation work?
2. Make the quick-and-dirty AI chart
3. Pick apart what went wrong
4. Treat the chart as a claim, not decoration
5. Make it narrative and opinionated
6. Build a repeatable workflow
7. Package what you learnt

About me

- I founded **Babbage Insight**, an AI analytics startup
- Earlier I was SVP of data at **Delhivery**
- I write a column for **Mint**. At some point someone called me "India's Nate Silver" (!)
- I blog at noenthuda.substack.com and artofdatascience.substack.com
- I used to write badvisualisations.tumblr.com and visualisations.substack.com
- Author of *Between the Buyer and the Seller*
- IIT Madras, IIM Bangalore
- These days I help teams move AI-for-data from clever demos to workflows people can trust

Keep open

- your dataset
- your LLM / coding tool
- this deck
- WhatsApp group

What makes a visualisation work?

Send to the WhatsApp group

- one visualisation you think is good
- one visualisation you think is bad
- don't say which is which

Note: Cmd+Tab to WhatsApp. Pick a few examples. Ask the room what they think. Maybe use StreamAlive if useful.

What did the room notice?

Note: build this slide live from the WhatsApp examples.

Look for:

- what people understood quickly
- where people disagreed
- what got overpraised because it looked nice
- what looked ugly but answered the question
- what was misleading without technically being wrong

Let's start badly

Upload your dataset to your LLM and paste this:

```
I have uploaded a dataset.

My question is: [paste your question here]

Make one chart that answers this question.
Choose the chart type yourself.
Use the data as given.
Add a title and any labels needed to understand the chart.
If you need to make assumptions, state them briefly.
```

Save the first output. Don't fix it yet.

Send the first chart to WhatsApp

No explanations yet.

Let other people react to the chart before you defend it.

Note: review a few. Ask people to comment in the group. Capture recurring failures.

What usually goes wrong?

Note: fill from discussion.

Likely suspects:

- chart answers a different question
- comparison is missing
- aggregation is lazy
- denominator is wrong or absent
- labels are technically correct but useless
- title describes the data, not the point
- visual form looks plausible but tests nothing
- caveat is hidden in prose nobody reads

LLMs are not magic

Before asking for a chart, we need to know:

- what data we have
- who the reader is
- what question we are asking
- what story might be true
- why anyone should care

No story, no chart. Only decoration.

Data visualisation follows the scientific method

A chart is not "make data pretty".

It is a small act of inquiry.

The basic loop is:

```
question → hypothesis → evidence → visual test → revise → communicate
```

The chart is where the claim meets the evidence.

What makes a good chart?

Kaiser Fung has a useful test.

A good visualisation needs three things to line up:

1. **Question** - what is this chart trying to answer?
2. **Data** - is this the right evidence for that question?
3. **Visual** - does the form make the answer easy to see?

If one corner is weak, the chart is weak.

Use the trifecta as a diagnostic

Common failures:

- good visual, wrong question → attractive but irrelevant
- good question, wrong data → persuasive but unsupported
- good data, wrong visual → true but hard to see

For AI charts, add one more check:

Can I review how it got here?

Start with the question

Weak:

```
Plot sales by month.
```

Better:

```
Did sales growth improve after the pricing change?
```

The second question gives the chart a job.

Without a job, the chart becomes wallpaper.

Say what you expect

Before plotting, write the hypothesis.

For example:

- urban districts recovered faster than rural ones
- the policy change created a visible break in trend
- most growth came from a few large customers
- the average hides widening inequality

If you cannot state the hypothesis, you probably don't yet know what to draw.

Treat the data as evidence

Before choosing the chart, inspect the evidence.

Ask:

- what is one row?
- what is the denominator?
- what is missing?
- which time period matters?
- what is the fair comparison?
- is this a count, rate, share, index, or rank?

Many bad charts are not design failures.

They are evidence failures.

The chart type is the test

Different claims need different tests:

- change over time → line chart
- ranking → ordered bar chart
- distribution → histogram / box / density
- relationship → scatterplot
- many groups over time → small multiples
- actual vs target → bullet / range chart
- before vs after → slopegraph or indexed line

This is just my way of visualisation. Yours might vary.
You build your own visual language

Control for obvious traps

Ask: what else could explain this pattern?

In chart terms:

- use rates when populations differ
- add a baseline, peer group, target, or earlier period
- don't use dramatic axes for tiny changes
- don't mix unlike groups and then average them
- show sample size or uncertainty when it matters

A chart without context is an experiment without controls.

What needs to survive scrutiny?

For your chart, write down:

- claim
- evidence
- comparison
- caveat
- easiest way to misread it

Charts need to be narrative and opinionated

A narrative chart should be easy to read and hard to misread.

It should have:

- one main point
- a clear comparison
- enough labels to avoid guessing
- annotations where the eye should stop
- caveats where the reader might overclaim
- a reading path

The basic rule

If the reader has to hunt for the point, the chart has failed.

Build the workflow

The rough sequence:

1. import / clean data
2. understand the rows and columns
3. write candidate hypotheses
4. check which ones survive
5. pick the main story
6. choose the visual form
7. specify labels, colour, and annotation
8. make the chart
9. critique it
10. revise once or twice

The model can help with the middle bits.

Divide the work

Give the model small jobs:

- profile the data
- suggest possible stories
- challenge the claim
- draft labels and annotations
- check the visual choice
- list ways the chart could mislead

Do not outsource taste or judgment.

Prompt: possible stories

```
Inspect the data using my context.
Do not make a chart yet.

Return 5 possible stories.
For each one, give:
- claim
- evidence
- comparison
- chart form
- confidence
- misleading risk

Rank them and recommend one to visualise.
```

Prompt: visual brief

```
Turn the chosen story into a visual brief.

Include:
- main claim
- audience
- evidence columns
- filters / aggregation
- comparison baseline
- chart form
- labels / annotations / colours
- caveats
- acceptance criteria
```

Prompt: make the chart

```
Using the context, taste rules, story, and visual brief,
make one static presentation chart.

Rules:
- one main claim
- visible comparison
- direct labels where possible
- no causal language unless the evidence supports it
- include source and caveat
```

Prompt: review the chart

```
Critique the rendered chart.

Check:
- analytical problems
- visual problems
- misleading readings
- taste-rule violations
- missing caveats

Then propose one revision.
```

Regenerate / refine

Don't keep patching forever.

If the thread becomes messy:

1. summarise context + story + brief + critique
2. start a fresh run
3. generate again

A clean restart is often faster than slow repair.

From chart to system

Production does not mean "ask ChatGPT nicely".

It means someone has decided:

- what the brief contains
- what the standards are
- what gets checked
- who owns the output
- when it refreshes
- when a human must intervene

Standardise this

If 50 people used this next month, what should be:

- standardised?
- banned?
- reviewed?
- escalated?

That is where the agent actually begins.

Package the workflow

Save the useful artefacts:

```
context.md
taste.md
story_candidates.md
visual_brief.md
chart_review.md
next_run.md
```

The chart is one run.

The workflow is the reusable thing.

Share-out

Show:

1. first AI chart
2. what failed
3. revised brief or chart
4. one rule you would reuse

Close

AI makes chart-making cheap.

Judgment makes charts worth looking at.

story + evidence + taste + review